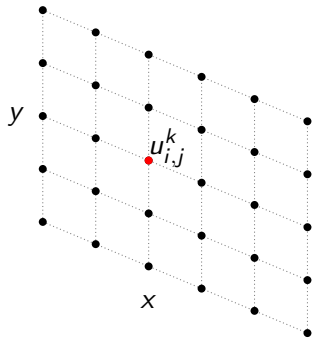
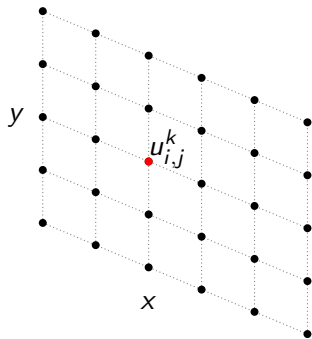


Discretize the Domain



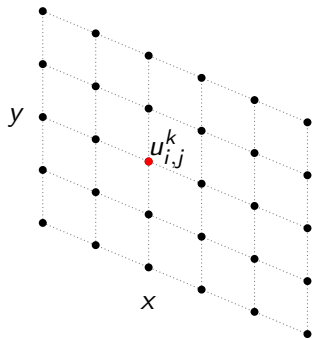
Discretize the Domain

- $u_{i,j}^k$ represents the concentration $u(x_i, y_j, t_k)$



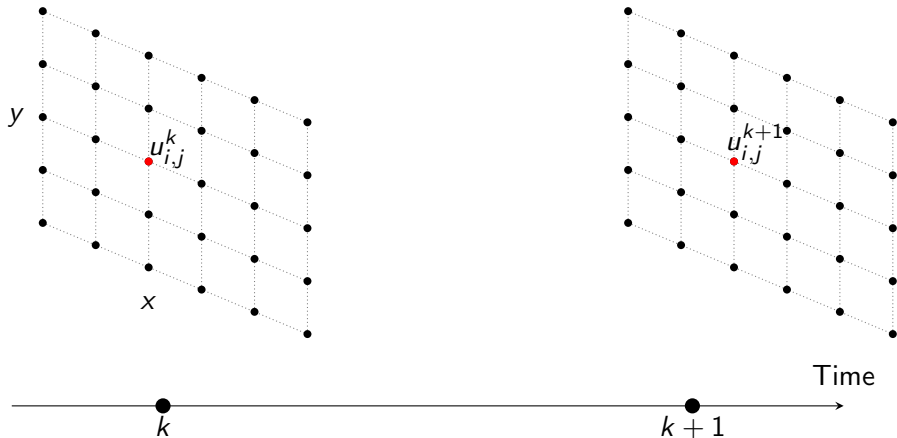
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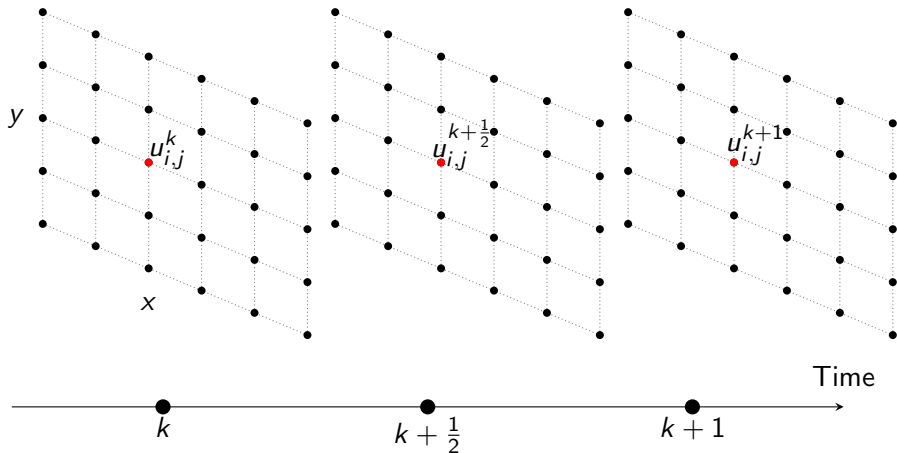
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Simulationg point-like sinks and sources

$$\frac{\partial u}{\partial t} = D \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) + \sum_{b=1}^N K_{out}(c_b) \delta(r_b - r) - K_{in}(u) \sum_{b=1}^N \delta(r_b - r)$$

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$$\frac{u_{i,j}^{k+\frac{1}{2}} - u_{i,j}^k}{\Delta t/2} = D \left(\frac{u_{i+1,j}^{k+\frac{1}{2}} - 2u_{i,j}^{k+\frac{1}{2}} + u_{i-1,j}^{k+\frac{1}{2}}}{\Delta x^2} + \frac{u_{i,j+1}^k - 2u_{i,j}^k + u_{i,j-1}^k}{\Delta y^2} \right) + S^k$$

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And the second half-step (x explicit, y implicit) using the following scheme:

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Where S is the sources and sinks term:

$$S^k = \sum_{b=1}^N K_{out}(c_b) \delta_\varepsilon(r_b - r_{i,j}) - K_{in}(u_{i,j}^k) \sum_{b=1}^N \delta_\varepsilon(r_b - r_{i,j})$$

Let's start with the first half-step:

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To simplify, we notice that $\Delta x = \Delta y$, and we can define $\alpha = \frac{D\Delta t}{2\Delta x^2}$ to get:

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Now we can rearrange the equation to isolate the unknown terms $u_{i,j}^{k+\frac{1}{2}}$ on the left-hand side:

$$-\alpha u_{i-1,j}^{k+\frac{1}{2}} + (1 + 2\alpha)u_{i,j}^{k+\frac{1}{2}} - \alpha u_{i+1,j}^{k+\frac{1}{2}} = \alpha u_{i,j-1}^k + (1 - 2\alpha)u_{i,j}^k + \alpha u_{i,j+1}^k + \frac{\Delta t}{2} S^k$$

We know all the terms in right-hand side of the equation (blue), but we need to compute the left-hand side (red).

First half-step (x implicit, y explicit)

$$-\alpha u_{i-1,j}^{k+\frac{1}{2}} + (1+2\alpha)u_{i,j}^{k+\frac{1}{2}} - \alpha u_{i+1,j}^{k+\frac{1}{2}} = \alpha u_{i,j-1}^k + (1-2\alpha)u_{i,j}^k + \alpha u_{i,j+1}^k + \frac{\Delta t}{2} S^k$$

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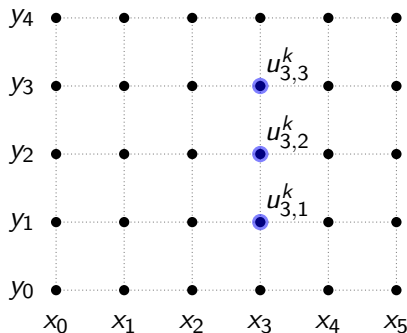
$$-\alpha u_{i-1,j}^{k+\frac{1}{2}} + (1+2\alpha)u_{i,j}^{k+\frac{1}{2}} - \alpha u_{i+1,j}^{k+\frac{1}{2}} = \alpha u_{i,j-1}^k + (1-2\alpha)u_{i,j}^k + \alpha u_{i,j+1}^k + \frac{\Delta t}{2} S^k$$

Let's look around $i = 3$ and $j = 2$:

First half-step (x implicit, y explicit)

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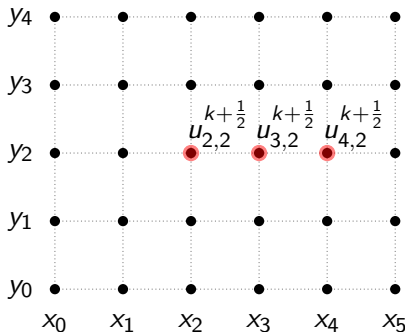
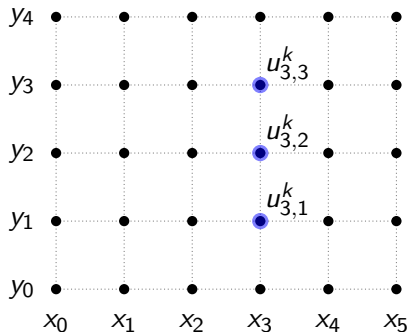
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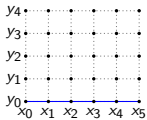
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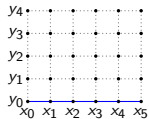
Let's fix $j = 0$ and write the equations for $0 \leq i \leq 5$.

k

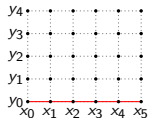


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k



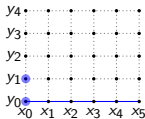
$k + \frac{1}{2}$



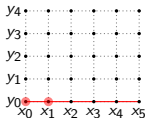
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$$-\alpha u_{-1,0}^{k+\frac{1}{2}} + (1+2\alpha)u_{0,0}^{k+\frac{1}{2}} - \alpha u_{1,0}^{k+\frac{1}{2}} = \alpha u_{0,-1}^k + (1-2\alpha)u_{0,0}^k + \alpha u_{0,1}^k + \frac{\Delta t}{2} S^k$$

k



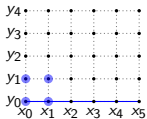
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 \end{aligned}$$

k



$k + \frac{1}{2}$



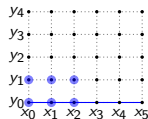
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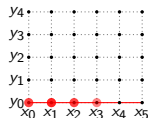
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k



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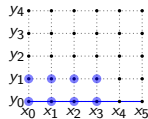
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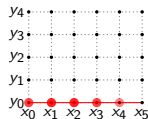
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$$-\alpha u_{2,0}^{k+\frac{1}{2}} + (1+2\alpha)u_{3,0}^{k+\frac{1}{2}} - \alpha u_{4,0}^{k+\frac{1}{2}} = \alpha u_{3,-1}^k + (1-2\alpha)u_{3,0}^k + \alpha u_{3,1}^k + \frac{\Delta t}{2} S^k$$

k

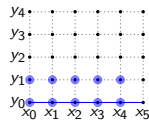


$k + \frac{1}{2}$



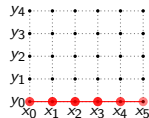
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k



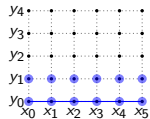
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 \end{aligned}$$

$k + \frac{1}{2}$



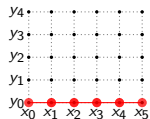
Let's fix $j = 0$ and write the equations for $0 \leq i \leq 5$.

k



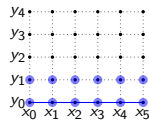
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 -\alpha u_{4,0}^{k+\frac{1}{2}} + (1+2\alpha) u_{5,0}^{k+\frac{1}{2}} - \alpha u_{6,0}^{k+\frac{1}{2}} &= \alpha u_{5,-1}^k + (1-2\alpha) u_{5,0}^k + \alpha u_{5,1}^k + \frac{\Delta t}{2} S^k
 \end{aligned}$$

$k + \frac{1}{2}$



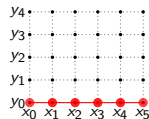
Let's fix $j = 0$ and write the equations for $0 \leq i \leq 5$.

k



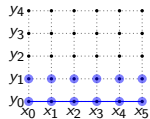
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 -\alpha u_{0,0}^{k+\frac{1}{2}} + (1+2\alpha) u_{1,0}^{k+\frac{1}{2}} - \alpha u_{2,0}^{k+\frac{1}{2}} &= \alpha u_{1,-1}^k + (1-2\alpha) u_{1,0}^k + \alpha u_{1,1}^k + \frac{\Delta t}{2} S^k \\
 -\alpha u_{1,0}^{k+\frac{1}{2}} + (1+2\alpha) u_{2,0}^{k+\frac{1}{2}} - \alpha u_{3,0}^{k+\frac{1}{2}} &= \alpha u_{2,-1}^k + (1-2\alpha) u_{2,0}^k + \alpha u_{2,1}^k + \frac{\Delta t}{2} S^k \\
 -\alpha u_{2,0}^{k+\frac{1}{2}} + (1+2\alpha) u_{3,0}^{k+\frac{1}{2}} - \alpha u_{4,0}^{k+\frac{1}{2}} &= \alpha u_{3,-1}^k + (1-2\alpha) u_{3,0}^k + \alpha u_{3,1}^k + \frac{\Delta t}{2} S^k \\
 -\alpha u_{3,0}^{k+\frac{1}{2}} + (1+2\alpha) u_{4,0}^{k+\frac{1}{2}} - \alpha u_{5,0}^{k+\frac{1}{2}} &= \alpha u_{4,-1}^k + (1-2\alpha) u_{4,0}^k + \alpha u_{4,1}^k + \frac{\Delta t}{2} S^k \\
 -\alpha u_{4,0}^{k+\frac{1}{2}} + (1+2\alpha) u_{5,0}^{k+\frac{1}{2}} - \alpha u_{6,0}^{k+\frac{1}{2}} &= \alpha u_{5,-1}^k + (1-2\alpha) u_{5,0}^k + \alpha u_{5,1}^k + \frac{\Delta t}{2} S^k
 \end{aligned}$$

$k + \frac{1}{2}$



Let's fix $j = 0$ and write the equations for $0 \leq i \leq 5$.

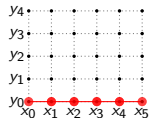
k



$$\begin{aligned}
 -\alpha u_{-1,0}^{k+\frac{1}{2}} + (1+2\alpha) u_{0,0}^{k+\frac{1}{2}} - \alpha u_{1,0}^{k+\frac{1}{2}} &= \alpha u_{0,-1}^k + (1-2\alpha) u_{0,0}^k + \alpha u_{0,1}^k + \frac{\Delta t}{2} S^k \\
 -\alpha u_{0,0}^{k+\frac{1}{2}} + (1+2\alpha) u_{1,0}^{k+\frac{1}{2}} - \alpha u_{2,0}^{k+\frac{1}{2}} &= \alpha u_{1,-1}^k + (1-2\alpha) u_{1,0}^k + \alpha u_{1,1}^k + \frac{\Delta t}{2} S^k \\
 -\alpha u_{1,0}^{k+\frac{1}{2}} + (1+2\alpha) u_{2,0}^{k+\frac{1}{2}} - \alpha u_{3,0}^{k+\frac{1}{2}} &= \alpha u_{2,-1}^k + (1-2\alpha) u_{2,0}^k + \alpha u_{2,1}^k + \frac{\Delta t}{2} S^k \\
 -\alpha u_{2,0}^{k+\frac{1}{2}} + (1+2\alpha) u_{3,0}^{k+\frac{1}{2}} - \alpha u_{4,0}^{k+\frac{1}{2}} &= \alpha u_{3,-1}^k + (1-2\alpha) u_{3,0}^k + \alpha u_{3,1}^k + \frac{\Delta t}{2} S^k \\
 -\alpha u_{3,0}^{k+\frac{1}{2}} + (1+2\alpha) u_{4,0}^{k+\frac{1}{2}} - \alpha u_{5,0}^{k+\frac{1}{2}} &= \alpha u_{4,-1}^k + (1-2\alpha) u_{4,0}^k + \alpha u_{4,1}^k + \frac{\Delta t}{2} S^k \\
 -\alpha u_{4,0}^{k+\frac{1}{2}} + (1+2\alpha) u_{5,0}^{k+\frac{1}{2}} - \alpha u_{6,0}^{k+\frac{1}{2}} &= \alpha u_{5,-1}^k + (1-2\alpha) u_{5,0}^k + \alpha u_{5,1}^k + \frac{\Delta t}{2} S^k
 \end{aligned}$$

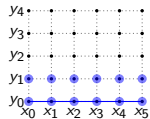
Which can be written as a matrix:

$k + \frac{1}{2}$



Let's fix $j = 0$ and write the equations for $0 \leq i \leq 5$.

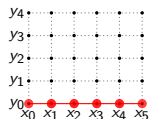
k



$$\begin{aligned}
 -\alpha u_{-1,0}^{k+\frac{1}{2}} + (1+2\alpha) u_{0,0}^{k+\frac{1}{2}} - \alpha u_{1,0}^{k+\frac{1}{2}} &= \alpha u_{0,-1}^k + (1-2\alpha) u_{0,0}^k + \alpha u_{0,1}^k + \frac{\Delta t}{2} S^k \\
 -\alpha u_{0,0}^{k+\frac{1}{2}} + (1+2\alpha) u_{1,0}^{k+\frac{1}{2}} - \alpha u_{2,0}^{k+\frac{1}{2}} &= \alpha u_{1,-1}^k + (1-2\alpha) u_{1,0}^k + \alpha u_{1,1}^k + \frac{\Delta t}{2} S^k \\
 -\alpha u_{1,0}^{k+\frac{1}{2}} + (1+2\alpha) u_{2,0}^{k+\frac{1}{2}} - \alpha u_{3,0}^{k+\frac{1}{2}} &= \alpha u_{2,-1}^k + (1-2\alpha) u_{2,0}^k + \alpha u_{2,1}^k + \frac{\Delta t}{2} S^k \\
 -\alpha u_{2,0}^{k+\frac{1}{2}} + (1+2\alpha) u_{3,0}^{k+\frac{1}{2}} - \alpha u_{4,0}^{k+\frac{1}{2}} &= \alpha u_{3,-1}^k + (1-2\alpha) u_{3,0}^k + \alpha u_{3,1}^k + \frac{\Delta t}{2} S^k \\
 -\alpha u_{3,0}^{k+\frac{1}{2}} + (1+2\alpha) u_{4,0}^{k+\frac{1}{2}} - \alpha u_{5,0}^{k+\frac{1}{2}} &= \alpha u_{4,-1}^k + (1-2\alpha) u_{4,0}^k + \alpha u_{4,1}^k + \frac{\Delta t}{2} S^k \\
 -\alpha u_{4,0}^{k+\frac{1}{2}} + (1+2\alpha) u_{5,0}^{k+\frac{1}{2}} - \alpha u_{6,0}^{k+\frac{1}{2}} &= \alpha u_{5,-1}^k + (1-2\alpha) u_{5,0}^k + \alpha u_{5,1}^k + \frac{\Delta t}{2} S^k
 \end{aligned}$$

Which can be written as a matrix:

$k + \frac{1}{2}$



$$\begin{bmatrix}
 -\alpha & 1+2\alpha & -\alpha & 0 & 0 & 0 & 0 & 0 \\
 0 & -\alpha & 1+2\alpha & -\alpha & 0 & 0 & 0 & 0 \\
 0 & 0 & -\alpha & 1+2\alpha & -\alpha & 0 & 0 & 0 \\
 0 & 0 & 0 & -\alpha & 1+2\alpha & -\alpha & 0 & 0 \\
 0 & 0 & 0 & 0 & -\alpha & 1+2\alpha & -\alpha & 0 \\
 0 & 0 & 0 & 0 & 0 & -\alpha & 1+2\alpha & -\alpha
 \end{bmatrix}
 \begin{bmatrix}
 u_{-1,0}^{k+\frac{1}{2}} \\
 u_{0,0}^{k+\frac{1}{2}} \\
 u_{1,0}^{k+\frac{1}{2}} \\
 u_{2,0}^{k+\frac{1}{2}} \\
 u_{3,0}^{k+\frac{1}{2}} \\
 u_{4,0}^{k+\frac{1}{2}} \\
 u_{5,0}^{k+\frac{1}{2}} \\
 u_{6,0}^{k+\frac{1}{2}}
 \end{bmatrix}
 =
 \begin{bmatrix}
 b_{0,0} \\
 b_{1,0} \\
 b_{2,0} \\
 b_{3,0} \\
 b_{4,0} \\
 b_{5,0}
 \end{bmatrix}$$

$$\begin{bmatrix}
 -\alpha & 1+2\alpha & -\alpha & 0 & 0 & 0 & 0 & 0 \\
 0 & -\alpha & 1+2\alpha & -\alpha & 0 & 0 & 0 & 0 \\
 0 & 0 & -\alpha & 1+2\alpha & -\alpha & 0 & 0 & 0 \\
 0 & 0 & 0 & -\alpha & 1+2\alpha & -\alpha & 0 & 0 \\
 0 & 0 & 0 & 0 & -\alpha & 1+2\alpha & -\alpha & 0 \\
 0 & 0 & 0 & 0 & 0 & -\alpha & 1+2\alpha & -\alpha
 \end{bmatrix}
 \begin{bmatrix}
 u_{-1,0}^{k+\frac{1}{2}} \\
 u_{0,0}^{k+\frac{1}{2}} \\
 u_{1,0}^{k+\frac{1}{2}} \\
 u_{2,0}^{k+\frac{1}{2}} \\
 u_{3,0}^{k+\frac{1}{2}} \\
 u_{4,0}^{k+\frac{1}{2}} \\
 u_{5,0}^{k+\frac{1}{2}} \\
 u_{6,0}^{k+\frac{1}{2}}
 \end{bmatrix}
 =
 \begin{bmatrix}
 b_{0,0} \\
 b_{1,0} \\
 b_{2,0} \\
 b_{3,0} \\
 b_{4,0} \\
 b_{5,0}
 \end{bmatrix}$$

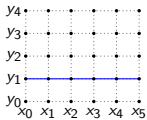
$$\begin{bmatrix}
 -\alpha & 1+2\alpha & -\alpha & 0 & 0 & 0 & 0 & 0 \\
 0 & -\alpha & 1+2\alpha & -\alpha & 0 & 0 & 0 & 0 \\
 0 & 0 & -\alpha & 1+2\alpha & -\alpha & 0 & 0 & 0 \\
 0 & 0 & 0 & -\alpha & 1+2\alpha & -\alpha & 0 & 0 \\
 0 & 0 & 0 & 0 & -\alpha & 1+2\alpha & -\alpha & 0 \\
 0 & 0 & 0 & 0 & 0 & -\alpha & 1+2\alpha & -\alpha
 \end{bmatrix}
 \begin{bmatrix}
 u_{-1,0}^{k+\frac{1}{2}} \\
 u_{0,0}^{k+\frac{1}{2}} \\
 u_{1,0}^{k+\frac{1}{2}} \\
 u_{2,0}^{k+\frac{1}{2}} \\
 u_{3,0}^{k+\frac{1}{2}} \\
 u_{4,0}^{k+\frac{1}{2}} \\
 u_{5,0}^{k+\frac{1}{2}} \\
 u_{6,0}^{k+\frac{1}{2}}
 \end{bmatrix}
 =
 \begin{bmatrix}
 b_{0,0} \\
 b_{1,0} \\
 b_{2,0} \\
 b_{3,0} \\
 b_{4,0} \\
 b_{5,0}
 \end{bmatrix}$$

We know the values on the boundaries, so we don't need to include them in the matrix.

$$\begin{bmatrix}
 1+2\alpha & -\alpha & 0 & 0 & 0 & 0 \\
 -\alpha & 1+2\alpha & -\alpha & 0 & 0 & 0 \\
 0 & -\alpha & 1+2\alpha & -\alpha & 0 & 0 \\
 0 & 0 & -\alpha & 1+2\alpha & -\alpha & 0 \\
 0 & 0 & 0 & -\alpha & 1+2\alpha & -\alpha \\
 0 & 0 & 0 & 0 & -\alpha & 1+2\alpha
 \end{bmatrix}
 \begin{bmatrix}
 u_{0,0}^{k+\frac{1}{2}} \\
 u_{1,0}^{k+\frac{1}{2}} \\
 u_{2,0}^{k+\frac{1}{2}} \\
 u_{3,0}^{k+\frac{1}{2}} \\
 u_{4,0}^{k+\frac{1}{2}} \\
 u_{5,0}^{k+\frac{1}{2}}
 \end{bmatrix}
 =
 \begin{bmatrix}
 b_{0,0} \\
 b_{1,0} \\
 b_{2,0} \\
 b_{3,0} \\
 b_{4,0} \\
 b_{5,0}
 \end{bmatrix}$$

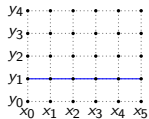
Let's fix $j = 1$ and write the equations for $0 \leq i \leq 5$.

k

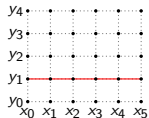


Let's fix $j = 1$ and write the equations for $0 \leq i \leq 5$.

k



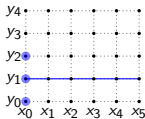
$k + \frac{1}{2}$



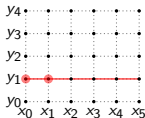
Let's fix $j = 1$ and write the equations for $0 \leq i \leq 5$.

$$-\alpha u_{-1,1}^{k+\frac{1}{2}} + (1+2\alpha)u_{0,1}^{k+\frac{1}{2}} - \alpha u_{1,1}^{k+\frac{1}{2}} = \alpha u_{0,0}^k + (1-2\alpha)u_{0,1}^k + \alpha u_{0,2}^k + \frac{\Delta t}{2} S^k$$

k



$k + \frac{1}{2}$

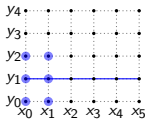


Let's fix $j = 1$ and write the equations for $0 \leq i \leq 5$.

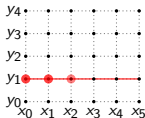
$$-\alpha u_{-1,1}^{k+\frac{1}{2}} + (1+2\alpha)u_{0,1}^{k+\frac{1}{2}} - \alpha u_{1,1}^{k+\frac{1}{2}} = \alpha u_{0,0}^k + (1-2\alpha)u_{0,1}^k + \alpha u_{0,2}^k + \frac{\Delta t}{2} S^k$$

$$-\alpha u_{0,1}^{k+\frac{1}{2}} + (1+2\alpha)u_{1,1}^{k+\frac{1}{2}} - \alpha u_{2,1}^{k+\frac{1}{2}} = \alpha u_{1,0}^k + (1-2\alpha)u_{1,1}^k + \alpha u_{1,2}^k + \frac{\Delta t}{2} S^k$$

k



$k + \frac{1}{2}$



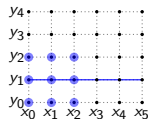
Let's fix $j = 1$ and write the equations for $0 \leq i \leq 5$.

$$-\alpha u_{-1,1}^{k+\frac{1}{2}} + (1+2\alpha)u_{0,1}^{k+\frac{1}{2}} - \alpha u_{1,1}^{k+\frac{1}{2}} = \alpha u_{0,0}^k + (1-2\alpha)u_{0,1}^k + \alpha u_{0,2}^k + \frac{\Delta t}{2} S^k$$

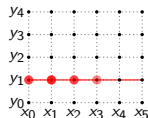
$$-\alpha u_{0,1}^{k+\frac{1}{2}} + (1+2\alpha)u_{1,1}^{k+\frac{1}{2}} - \alpha u_{2,1}^{k+\frac{1}{2}} = \alpha u_{1,0}^k + (1-2\alpha)u_{1,1}^k + \alpha u_{1,2}^k + \frac{\Delta t}{2} S^k$$

$$-\alpha u_{1,1}^{k+\frac{1}{2}} + (1+2\alpha)u_{2,1}^{k+\frac{1}{2}} - \alpha u_{3,1}^{k+\frac{1}{2}} = \alpha u_{2,0}^k + (1-2\alpha)u_{2,1}^k + \alpha u_{2,2}^k + \frac{\Delta t}{2} S^k$$

k



$k + \frac{1}{2}$



Let's fix $j = 1$ and write the equations for $0 \leq i \leq 5$.

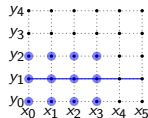
$$-\alpha u_{-1,1}^{k+\frac{1}{2}} + (1+2\alpha)u_{0,1}^{k+\frac{1}{2}} - \alpha u_{1,1}^{k+\frac{1}{2}} = \alpha u_{0,0}^k + (1-2\alpha)u_{0,1}^k + \alpha u_{0,2}^k + \frac{\Delta t}{2} S^k$$

$$-\alpha u_{0,1}^{k+\frac{1}{2}} + (1+2\alpha)u_{1,1}^{k+\frac{1}{2}} - \alpha u_{2,1}^{k+\frac{1}{2}} = \alpha u_{1,0}^k + (1-2\alpha)u_{1,1}^k + \alpha u_{1,2}^k + \frac{\Delta t}{2} S^k$$

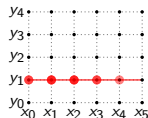
$$-\alpha u_{1,1}^{k+\frac{1}{2}} + (1+2\alpha)u_{2,1}^{k+\frac{1}{2}} - \alpha u_{3,1}^{k+\frac{1}{2}} = \alpha u_{2,0}^k + (1-2\alpha)u_{2,1}^k + \alpha u_{2,2}^k + \frac{\Delta t}{2} S^k$$

$$-\alpha u_{2,1}^{k+\frac{1}{2}} + (1+2\alpha)u_{3,1}^{k+\frac{1}{2}} - \alpha u_{4,1}^{k+\frac{1}{2}} = \alpha u_{3,0}^k + (1-2\alpha)u_{3,1}^k + \alpha u_{3,2}^k + \frac{\Delta t}{2} S^k$$

k

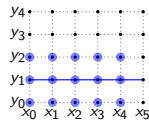


$k + \frac{1}{2}$



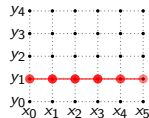
Let's fix $j = 1$ and write the equations for $0 \leq i \leq 5$.

k



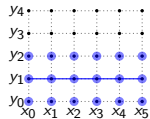
$$\begin{aligned}
 -\alpha u_{-1,1}^{k+\frac{1}{2}} + (1+2\alpha)u_{0,1}^{k+\frac{1}{2}} - \alpha u_{1,1}^{k+\frac{1}{2}} &= \alpha u_{0,0}^k + (1-2\alpha)u_{0,1}^k + \alpha u_{0,2}^k + \frac{\Delta t}{2} S^k \\
 -\alpha u_{0,1}^{k+\frac{1}{2}} + (1+2\alpha)u_{1,1}^{k+\frac{1}{2}} - \alpha u_{2,1}^{k+\frac{1}{2}} &= \alpha u_{1,0}^k + (1-2\alpha)u_{1,1}^k + \alpha u_{1,2}^k + \frac{\Delta t}{2} S^k \\
 -\alpha u_{1,1}^{k+\frac{1}{2}} + (1+2\alpha)u_{2,1}^{k+\frac{1}{2}} - \alpha u_{3,1}^{k+\frac{1}{2}} &= \alpha u_{2,0}^k + (1-2\alpha)u_{2,1}^k + \alpha u_{2,2}^k + \frac{\Delta t}{2} S^k \\
 -\alpha u_{2,1}^{k+\frac{1}{2}} + (1+2\alpha)u_{3,1}^{k+\frac{1}{2}} - \alpha u_{4,1}^{k+\frac{1}{2}} &= \alpha u_{3,0}^k + (1-2\alpha)u_{3,1}^k + \alpha u_{3,2}^k + \frac{\Delta t}{2} S^k \\
 -\alpha u_{3,1}^{k+\frac{1}{2}} + (1+2\alpha)u_{4,1}^{k+\frac{1}{2}} - \alpha u_{5,1}^{k+\frac{1}{2}} &= \alpha u_{4,0}^k + (1-2\alpha)u_{4,1}^k + \alpha u_{4,2}^k + \frac{\Delta t}{2} S^k
 \end{aligned}$$

$k + \frac{1}{2}$



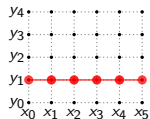
Let's fix $j = 1$ and write the equations for $0 \leq i \leq 5$.

k



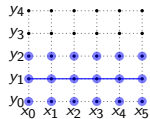
$$\begin{aligned}
 -\alpha u_{-1,1}^{k+\frac{1}{2}} + (1+2\alpha)u_{0,1}^{k+\frac{1}{2}} - \alpha u_{1,1}^{k+\frac{1}{2}} &= \alpha u_{0,0}^k + (1-2\alpha)u_{0,1}^k + \alpha u_{0,2}^k + \frac{\Delta t}{2} S^k \\
 -\alpha u_{0,1}^{k+\frac{1}{2}} + (1+2\alpha)u_{1,1}^{k+\frac{1}{2}} - \alpha u_{2,1}^{k+\frac{1}{2}} &= \alpha u_{1,0}^k + (1-2\alpha)u_{1,1}^k + \alpha u_{1,2}^k + \frac{\Delta t}{2} S^k \\
 -\alpha u_{1,1}^{k+\frac{1}{2}} + (1+2\alpha)u_{2,1}^{k+\frac{1}{2}} - \alpha u_{3,1}^{k+\frac{1}{2}} &= \alpha u_{2,0}^k + (1-2\alpha)u_{2,1}^k + \alpha u_{2,2}^k + \frac{\Delta t}{2} S^k \\
 -\alpha u_{2,1}^{k+\frac{1}{2}} + (1+2\alpha)u_{3,1}^{k+\frac{1}{2}} - \alpha u_{4,1}^{k+\frac{1}{2}} &= \alpha u_{3,0}^k + (1-2\alpha)u_{3,1}^k + \alpha u_{3,2}^k + \frac{\Delta t}{2} S^k \\
 -\alpha u_{3,1}^{k+\frac{1}{2}} + (1+2\alpha)u_{4,1}^{k+\frac{1}{2}} - \alpha u_{5,1}^{k+\frac{1}{2}} &= \alpha u_{4,0}^k + (1-2\alpha)u_{4,1}^k + \alpha u_{4,2}^k + \frac{\Delta t}{2} S^k \\
 -\alpha u_{4,1}^{k+\frac{1}{2}} + (1+2\alpha)u_{5,1}^{k+\frac{1}{2}} - \alpha u_{6,1}^{k+\frac{1}{2}} &= \alpha u_{5,0}^k + (1-2\alpha)u_{5,1}^k + \alpha u_{5,2}^k + \frac{\Delta t}{2} S^k
 \end{aligned}$$

$k + \frac{1}{2}$



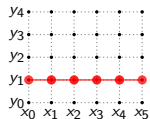
Let's fix $j = 1$ and write the equations for $0 \leq i \leq 5$.

k



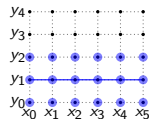
$$\begin{aligned}
 -\alpha u_{-1,1}^{k+\frac{1}{2}} + (1+2\alpha)u_{0,1}^{k+\frac{1}{2}} - \alpha u_{1,1}^{k+\frac{1}{2}} &= \alpha u_{0,0}^k + (1-2\alpha)u_{0,1}^k + \alpha u_{0,2}^k + \frac{\Delta t}{2} S^k \\
 -\alpha u_{0,1}^{k+\frac{1}{2}} + (1+2\alpha)u_{1,1}^{k+\frac{1}{2}} - \alpha u_{2,1}^{k+\frac{1}{2}} &= \alpha u_{1,0}^k + (1-2\alpha)u_{1,1}^k + \alpha u_{1,2}^k + \frac{\Delta t}{2} S^k \\
 -\alpha u_{1,1}^{k+\frac{1}{2}} + (1+2\alpha)u_{2,1}^{k+\frac{1}{2}} - \alpha u_{3,1}^{k+\frac{1}{2}} &= \alpha u_{2,0}^k + (1-2\alpha)u_{2,1}^k + \alpha u_{2,2}^k + \frac{\Delta t}{2} S^k \\
 -\alpha u_{2,1}^{k+\frac{1}{2}} + (1+2\alpha)u_{3,1}^{k+\frac{1}{2}} - \alpha u_{4,1}^{k+\frac{1}{2}} &= \alpha u_{3,0}^k + (1-2\alpha)u_{3,1}^k + \alpha u_{3,2}^k + \frac{\Delta t}{2} S^k \\
 -\alpha u_{3,1}^{k+\frac{1}{2}} + (1+2\alpha)u_{4,1}^{k+\frac{1}{2}} - \alpha u_{5,1}^{k+\frac{1}{2}} &= \alpha u_{4,0}^k + (1-2\alpha)u_{4,1}^k + \alpha u_{4,2}^k + \frac{\Delta t}{2} S^k \\
 -\alpha u_{4,1}^{k+\frac{1}{2}} + (1+2\alpha)u_{5,1}^{k+\frac{1}{2}} - \alpha u_{6,1}^{k+\frac{1}{2}} &= \alpha u_{5,0}^k + (1-2\alpha)u_{5,1}^k + \alpha u_{5,2}^k + \frac{\Delta t}{2} S^k
 \end{aligned}$$

$k + \frac{1}{2}$



Let's fix $j = 1$ and write the equations for $0 \leq i \leq 5$.

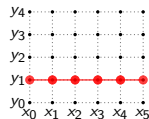
k



$$\begin{aligned}
 -\alpha u_{-1,1}^{k+\frac{1}{2}} + (1+2\alpha)u_{0,1}^{k+\frac{1}{2}} - \alpha u_{1,1}^{k+\frac{1}{2}} &= \alpha u_{0,0}^k + (1-2\alpha)u_{0,1}^k + \alpha u_{0,2}^k + \frac{\Delta t}{2} S^k \\
 -\alpha u_{0,1}^{k+\frac{1}{2}} + (1+2\alpha)u_{1,1}^{k+\frac{1}{2}} - \alpha u_{2,1}^{k+\frac{1}{2}} &= \alpha u_{1,0}^k + (1-2\alpha)u_{1,1}^k + \alpha u_{1,2}^k + \frac{\Delta t}{2} S^k \\
 -\alpha u_{1,1}^{k+\frac{1}{2}} + (1+2\alpha)u_{2,1}^{k+\frac{1}{2}} - \alpha u_{3,1}^{k+\frac{1}{2}} &= \alpha u_{2,0}^k + (1-2\alpha)u_{2,1}^k + \alpha u_{2,2}^k + \frac{\Delta t}{2} S^k \\
 -\alpha u_{2,1}^{k+\frac{1}{2}} + (1+2\alpha)u_{3,1}^{k+\frac{1}{2}} - \alpha u_{4,1}^{k+\frac{1}{2}} &= \alpha u_{3,0}^k + (1-2\alpha)u_{3,1}^k + \alpha u_{3,2}^k + \frac{\Delta t}{2} S^k \\
 -\alpha u_{3,1}^{k+\frac{1}{2}} + (1+2\alpha)u_{4,1}^{k+\frac{1}{2}} - \alpha u_{5,1}^{k+\frac{1}{2}} &= \alpha u_{4,0}^k + (1-2\alpha)u_{4,1}^k + \alpha u_{4,2}^k + \frac{\Delta t}{2} S^k \\
 -\alpha u_{4,1}^{k+\frac{1}{2}} + (1+2\alpha)u_{5,1}^{k+\frac{1}{2}} - \alpha u_{6,1}^{k+\frac{1}{2}} &= \alpha u_{5,0}^k + (1-2\alpha)u_{5,1}^k + \alpha u_{5,2}^k + \frac{\Delta t}{2} S^k
 \end{aligned}$$

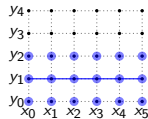
Which can be written as a matrix:

$k + \frac{1}{2}$



Let's fix $j = 1$ and write the equations for $0 \leq i \leq 5$.

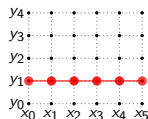
k



$$\begin{aligned}
 -\alpha u_{-1,1}^{k+\frac{1}{2}} + (1+2\alpha)u_{0,1}^{k+\frac{1}{2}} - \alpha u_{1,1}^{k+\frac{1}{2}} &= \alpha u_{0,0}^k + (1-2\alpha)u_{0,1}^k + \alpha u_{0,2}^k + \frac{\Delta t}{2} S^k \\
 -\alpha u_{0,1}^{k+\frac{1}{2}} + (1+2\alpha)u_{1,1}^{k+\frac{1}{2}} - \alpha u_{2,1}^{k+\frac{1}{2}} &= \alpha u_{1,0}^k + (1-2\alpha)u_{1,1}^k + \alpha u_{1,2}^k + \frac{\Delta t}{2} S^k \\
 -\alpha u_{1,1}^{k+\frac{1}{2}} + (1+2\alpha)u_{2,1}^{k+\frac{1}{2}} - \alpha u_{3,1}^{k+\frac{1}{2}} &= \alpha u_{2,0}^k + (1-2\alpha)u_{2,1}^k + \alpha u_{2,2}^k + \frac{\Delta t}{2} S^k \\
 -\alpha u_{2,1}^{k+\frac{1}{2}} + (1+2\alpha)u_{3,1}^{k+\frac{1}{2}} - \alpha u_{4,1}^{k+\frac{1}{2}} &= \alpha u_{3,0}^k + (1-2\alpha)u_{3,1}^k + \alpha u_{3,2}^k + \frac{\Delta t}{2} S^k \\
 -\alpha u_{3,1}^{k+\frac{1}{2}} + (1+2\alpha)u_{4,1}^{k+\frac{1}{2}} - \alpha u_{5,1}^{k+\frac{1}{2}} &= \alpha u_{4,0}^k + (1-2\alpha)u_{4,1}^k + \alpha u_{4,2}^k + \frac{\Delta t}{2} S^k \\
 -\alpha u_{4,1}^{k+\frac{1}{2}} + (1+2\alpha)u_{5,1}^{k+\frac{1}{2}} - \alpha u_{6,1}^{k+\frac{1}{2}} &= \alpha u_{5,0}^k + (1-2\alpha)u_{5,1}^k + \alpha u_{5,2}^k + \frac{\Delta t}{2} S^k
 \end{aligned}$$

Which can be written as a matrix:

$k + \frac{1}{2}$



$$\begin{bmatrix}
 -\alpha & 1+2\alpha & -\alpha & 0 & 0 & 0 & 0 & 0 \\
 0 & -\alpha & 1+2\alpha & -\alpha & 0 & 0 & 0 & 0 \\
 0 & 0 & -\alpha & 1+2\alpha & -\alpha & 0 & 0 & 0 \\
 0 & 0 & 0 & -\alpha & 1+2\alpha & -\alpha & 0 & 0 \\
 0 & 0 & 0 & 0 & -\alpha & 1+2\alpha & -\alpha & 0 \\
 0 & 0 & 0 & 0 & 0 & -\alpha & 1+2\alpha & -\alpha
 \end{bmatrix}
 \begin{bmatrix}
 u_{-1,1}^{k+\frac{1}{2}} \\
 u_{0,1}^{k+\frac{1}{2}} \\
 u_{1,1}^{k+\frac{1}{2}} \\
 u_{2,1}^{k+\frac{1}{2}} \\
 u_{3,1}^{k+\frac{1}{2}} \\
 u_{4,1}^{k+\frac{1}{2}} \\
 u_{5,1}^{k+\frac{1}{2}} \\
 u_{6,1}^{k+\frac{1}{2}}
 \end{bmatrix}
 =
 \begin{bmatrix}
 b_{0,1} \\
 b_{1,1} \\
 b_{2,1} \\
 b_{3,1} \\
 b_{4,1} \\
 b_{5,1}
 \end{bmatrix}$$

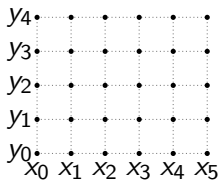
$$\begin{bmatrix}
 -\alpha & 1+2\alpha & -\alpha & 0 & 0 & 0 & 0 & 0 \\
 0 & -\alpha & 1+2\alpha & -\alpha & 0 & 0 & 0 & 0 \\
 0 & 0 & -\alpha & 1+2\alpha & -\alpha & 0 & 0 & 0 \\
 0 & 0 & 0 & -\alpha & 1+2\alpha & -\alpha & 0 & 0 \\
 0 & 0 & 0 & 0 & -\alpha & 1+2\alpha & -\alpha & 0 \\
 0 & 0 & 0 & 0 & 0 & -\alpha & 1+2\alpha & -\alpha
 \end{bmatrix}
 \begin{bmatrix}
 k+\frac{1}{2} \\
 u_{-1,1} \\
 k+\frac{1}{2} \\
 u_{0,1} \\
 k+\frac{1}{2} \\
 u_{1,1} \\
 k+\frac{1}{2} \\
 u_{2,1} \\
 k+\frac{1}{2} \\
 u_{3,1} \\
 k+\frac{1}{2} \\
 u_{4,1} \\
 k+\frac{1}{2} \\
 u_{5,1} \\
 k+\frac{1}{2} \\
 u_{6,1}
 \end{bmatrix}
 =
 \begin{bmatrix}
 b_{0,1} \\
 b_{1,1} \\
 b_{2,1} \\
 b_{3,1} \\
 b_{4,1} \\
 b_{5,1}
 \end{bmatrix}$$

$$\begin{bmatrix}
 -\alpha & 1+2\alpha & -\alpha & 0 & 0 & 0 & 0 & 0 \\
 0 & -\alpha & 1+2\alpha & -\alpha & 0 & 0 & 0 & 0 \\
 0 & 0 & -\alpha & 1+2\alpha & -\alpha & 0 & 0 & 0 \\
 0 & 0 & 0 & -\alpha & 1+2\alpha & -\alpha & 0 & 0 \\
 0 & 0 & 0 & 0 & -\alpha & 1+2\alpha & -\alpha & 0 \\
 0 & 0 & 0 & 0 & 0 & -\alpha & 1+2\alpha & -\alpha
 \end{bmatrix}
 \begin{bmatrix}
 u_{-1,1}^{k+\frac{1}{2}} \\
 u_{0,1}^{k+\frac{1}{2}} \\
 u_{1,1}^{k+\frac{1}{2}} \\
 u_{2,1}^{k+\frac{1}{2}} \\
 u_{3,1}^{k+\frac{1}{2}} \\
 u_{4,1}^{k+\frac{1}{2}} \\
 u_{5,1}^{k+\frac{1}{2}} \\
 u_{6,1}^{k+\frac{1}{2}}
 \end{bmatrix}
 =
 \begin{bmatrix}
 b_{0,1} \\
 b_{1,1} \\
 b_{2,1} \\
 b_{3,1} \\
 b_{4,1} \\
 b_{5,1}
 \end{bmatrix}$$

We know the values on the boundaries, so we don't need to include them in the matrix.

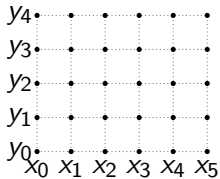
$$\begin{bmatrix}
 1+2\alpha & -\alpha & 0 & 0 & 0 & 0 \\
 -\alpha & 1+2\alpha & -\alpha & 0 & 0 & 0 \\
 0 & -\alpha & 1+2\alpha & -\alpha & 0 & 0 \\
 0 & 0 & -\alpha & 1+2\alpha & -\alpha & 0 \\
 0 & 0 & 0 & -\alpha & 1+2\alpha & -\alpha \\
 0 & 0 & 0 & 0 & -\alpha & 1+2\alpha
 \end{bmatrix}
 \begin{bmatrix}
 u_{0,1}^{k+\frac{1}{2}} \\
 u_{1,1}^{k+\frac{1}{2}} \\
 u_{2,1}^{k+\frac{1}{2}} \\
 u_{3,1}^{k+\frac{1}{2}} \\
 u_{4,1}^{k+\frac{1}{2}} \\
 u_{5,1}^{k+\frac{1}{2}}
 \end{bmatrix}
 =
 \begin{bmatrix}
 b_{0,1} \\
 b_{1,1} \\
 b_{2,1} \\
 b_{3,1} \\
 b_{4,1} \\
 b_{5,1}
 \end{bmatrix}$$

We get one system of equations for each j value.

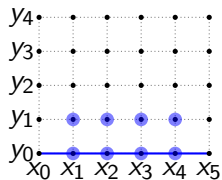


$\Delta t/2$

A vertical arrow pointing downwards, indicating a time step of $\Delta t/2$.

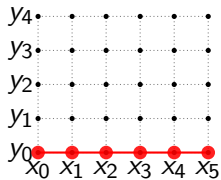


We get one system of equations for each j value.

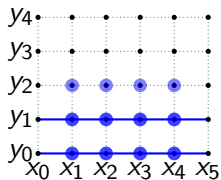


$$\begin{bmatrix} 1+2\alpha & -\alpha & 0 & 0 & 0 & 0 \\ -\alpha & 1+2\alpha & -\alpha & 0 & 0 & 0 \\ 0 & -\alpha & 1+2\alpha & -\alpha & 0 & 0 \\ 0 & 0 & -\alpha & 1+2\alpha & -\alpha & 0 \\ 0 & 0 & 0 & -\alpha & 1+2\alpha & -\alpha \\ 0 & 0 & 0 & 0 & -\alpha & 1+2\alpha \end{bmatrix} \begin{bmatrix} k+\frac{1}{2} \\ u_{0,0} \\ k+\frac{1}{2} \\ u_{1,0} \\ k+\frac{1}{2} \\ u_{2,0} \\ k+\frac{1}{2} \\ u_{3,0} \\ k+\frac{1}{2} \\ u_{4,0} \\ k+\frac{1}{2} \\ u_{5,0} \end{bmatrix} = \begin{bmatrix} b_{0,0} \\ b_{1,0} \\ b_{2,0} \\ b_{3,0} \\ b_{4,0} \\ b_{5,0} \end{bmatrix}$$

$\downarrow$
 $\Delta t/2$

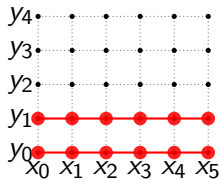


We get one system of equations for each j value.



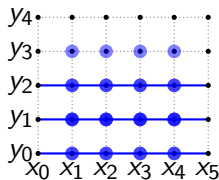
$$\begin{bmatrix} 1+2\alpha & -\alpha & 0 & 0 & 0 & 0 \\ -\alpha & 1+2\alpha & -\alpha & 0 & 0 & 0 \\ 0 & -\alpha & 1+2\alpha & -\alpha & 0 & 0 \\ 0 & 0 & -\alpha & 1+2\alpha & -\alpha & 0 \\ 0 & 0 & 0 & -\alpha & 1+2\alpha & -\alpha \\ 0 & 0 & 0 & 0 & -\alpha & 1+2\alpha \end{bmatrix} \begin{bmatrix} u_{0,0}^{k+\frac{1}{2}} \\ u_{1,0}^{k+\frac{1}{2}} \\ u_{2,0}^{k+\frac{1}{2}} \\ u_{3,0}^{k+\frac{1}{2}} \\ u_{4,0}^{k+\frac{1}{2}} \\ u_{5,0}^{k+\frac{1}{2}} \end{bmatrix} = \begin{bmatrix} b_{0,0} \\ b_{1,0} \\ b_{2,0} \\ b_{3,0} \\ b_{4,0} \\ b_{5,0} \end{bmatrix}$$

$\Delta t/2$



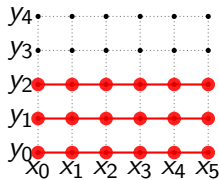
$$\begin{bmatrix} 1+2\alpha & -\alpha & 0 & 0 & 0 & 0 \\ -\alpha & 1+2\alpha & -\alpha & 0 & 0 & 0 \\ 0 & -\alpha & 1+2\alpha & -\alpha & 0 & 0 \\ 0 & 0 & -\alpha & 1+2\alpha & -\alpha & 0 \\ 0 & 0 & 0 & -\alpha & 1+2\alpha & -\alpha \\ 0 & 0 & 0 & 0 & -\alpha & 1+2\alpha \end{bmatrix} \begin{bmatrix} u_{0,1}^{k+\frac{1}{2}} \\ u_{1,1}^{k+\frac{1}{2}} \\ u_{2,1}^{k+\frac{1}{2}} \\ u_{3,1}^{k+\frac{1}{2}} \\ u_{4,1}^{k+\frac{1}{2}} \\ u_{5,1}^{k+\frac{1}{2}} \end{bmatrix} = \begin{bmatrix} b_{0,1} \\ b_{1,1} \\ b_{2,1} \\ b_{3,1} \\ b_{4,1} \\ b_{5,1} \end{bmatrix}$$

We get one system of equations for each j value.



$$\begin{bmatrix} 1+2\alpha & -\alpha & 0 & 0 & 0 & 0 \\ -\alpha & 1+2\alpha & -\alpha & 0 & 0 & 0 \\ 0 & -\alpha & 1+2\alpha & -\alpha & 0 & 0 \\ 0 & 0 & -\alpha & 1+2\alpha & -\alpha & 0 \\ 0 & 0 & 0 & -\alpha & 1+2\alpha & -\alpha \\ 0 & 0 & 0 & 0 & -\alpha & 1+2\alpha \end{bmatrix} \begin{bmatrix} k+\frac{1}{2} \\ u_{0,0} \\ k+\frac{1}{2} \\ u_{1,0} \\ k+\frac{1}{2} \\ u_{2,0} \\ k+\frac{1}{2} \\ u_{3,0} \\ k+\frac{1}{2} \\ u_{4,0} \\ k+\frac{1}{2} \\ u_{5,0} \end{bmatrix} = \begin{bmatrix} b_{0,0} \\ b_{1,0} \\ b_{2,0} \\ b_{3,0} \\ b_{4,0} \\ b_{5,0} \end{bmatrix}$$

$\Delta t/2$



$$\begin{bmatrix} 1+2\alpha & -\alpha & 0 & 0 & 0 & 0 \\ -\alpha & 1+2\alpha & -\alpha & 0 & 0 & 0 \\ 0 & -\alpha & 1+2\alpha & -\alpha & 0 & 0 \\ 0 & 0 & -\alpha & 1+2\alpha & -\alpha & 0 \\ 0 & 0 & 0 & -\alpha & 1+2\alpha & -\alpha \\ 0 & 0 & 0 & 0 & -\alpha & 1+2\alpha \end{bmatrix} \begin{bmatrix} k+\frac{1}{2} \\ u_{0,1} \\ k+\frac{1}{2} \\ u_{1,1} \\ k+\frac{1}{2} \\ u_{2,1} \\ k+\frac{1}{2} \\ u_{3,1} \\ k+\frac{1}{2} \\ u_{4,1} \\ k+\frac{1}{2} \\ u_{5,1} \end{bmatrix} = \begin{bmatrix} b_{0,1} \\ b_{1,1} \\ b_{2,1} \\ b_{3,1} \\ b_{4,1} \\ b_{5,1} \end{bmatrix}$$

$$\begin{bmatrix} 1+2\alpha & -\alpha & 0 & 0 & 0 & 0 \\ -\alpha & 1+2\alpha & -\alpha & 0 & 0 & 0 \\ 0 & -\alpha & 1+2\alpha & -\alpha & 0 & 0 \\ 0 & 0 & -\alpha & 1+2\alpha & -\alpha & 0 \\ 0 & 0 & 0 & -\alpha & 1+2\alpha & -\alpha \\ 0 & 0 & 0 & 0 & -\alpha & 1+2\alpha \end{bmatrix} \begin{bmatrix} k+\frac{1}{2} \\ u_{0,2} \\ k+\frac{1}{2} \\ u_{1,2} \\ k+\frac{1}{2} \\ u_{2,2} \\ k+\frac{1}{2} \\ u_{3,2} \\ k+\frac{1}{2} \\ u_{4,2} \\ k+\frac{1}{2} \\ u_{5,2} \end{bmatrix} = \begin{bmatrix} b_{0,2} \\ b_{1,2} \\ b_{2,2} \\ b_{3,2} \\ b_{4,2} \\ b_{5,2} \end{bmatrix}$$